MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. If $\sqrt{3}\tan 2\theta - 3 = 0$, where 2θ is an acute angle, then the value of θ is: [1]

(A) 15° (B) 30° (C) 45° (D) 60°

Q2. The value of $\cos 0^\circ \cdot \cos 1^\circ \cdot \cos 2^\circ \cdots \cos 90^\circ$ is: [1]

(A) 0 (B) 1 (C) -1 (D) $\frac{1}{2}$

Q3. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$ with $0^\circ < A + B \leq 90^\circ$, then B is: [1]

(A) 15° (B) 45° (C) 30° (D) 60°

Q4. The value of $\frac{\sin^2 30^\circ + 4\cot^2 45^\circ - \sec^2 60^\circ}{\csc^2 30^\circ}$ is: [1]

(A) $\frac{1}{16}$ (B) $\frac{1}{4}$ (C) $\frac{3}{16}$ (D) 1

Q5. **Assertion (A):** If $\cos A + \cos^2 A = 1$, then $\sin^2 A + \sin^4 A = 1$.
Reason (R): $\cos A = 1 - \cos^2 A = \sin^2 A$, hence $\sin^4 A = \cos^2 A$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. If $\sec\theta + \tan\theta = 7$, find the value of $\sec\theta - \tan\theta$ and hence calculate $\cos\theta$. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. If $x = a\sec\theta + b\tan\theta$ and $y = a\tan\theta + b\sec\theta$, prove that $x^2 - y^2 = a^2 - b^2$. [3]

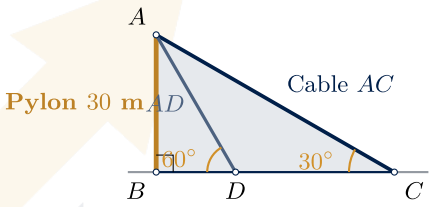
Q8. If $\sin\theta + \sin^2\theta + \sin^3\theta = 1$, prove that $\cos^6\theta - 4\cos^4\theta + 8\cos^2\theta = 4$. [3]

Q9. If $\sec\theta + \tan\theta = p$, prove that $\frac{p^2-1}{p^2+1} = \sin\theta$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Cable-Stayed Suspension Bridge Structural Guy Wires

A cable-stayed pedestrian bridge features a central vertical pylon AB of height 30 m standing perpendicular to the horizontal roadway BC. A steel stay cable AC connects the top of the pylon A to an anchor block C on the bridge deck. The cable makes an angle of $\theta = 30°$ with the horizontal roadway.



Based on the above geometric model, answer the following questions:

(i) Determine the length of the steel stay cable AC. [1]

(ii) Determine the distance BC from the base of the pylon to the anchor block. [1]

(iii) If a secondary cable AD is anchored at point D between B and C such that $\angle ADB = 60°$, find the length of cable AD and the distance CD between the two anchor points. [2]

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